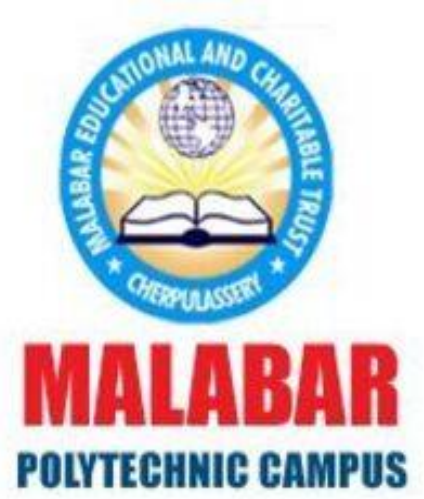
LOGIC SYSTEM

DESIGN LAB

MANUAL

3349



DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

MALABAR POLYTECHNIC CAMPUS
CHERPULASSERY,PALAKKAD

DEPARTMENT O F ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

LAB CYCLE FOR LOGIC SYSTEM DESIGN LAB

3349: Logic System Design Lab

CYCLE 1

1)Basic Logic Gates
2) De-Morgans theorem
3)Implement basic gates using universal gates
4)Half Adder& Half Substractor-Basic Gates
5) Half Adder &Half Substractor-Universal Gates
6) Mux and Demux

CYCLE 2

1)Realisation of D and T flipflops
2)Realisation of SR AND JK flipflops
3)Decade counter using IC 7490
4)Use of MUX IC 74153,Decoder/Demux IC 74139

EXPERIMENT 1

LOGIC GATES

Aim:-

Introduction to digital electronics lab- nomenclature of digital ICs, specifications, study of the data sheet, concept of Vcc and ground, verification of the truth tables of logic gates using TTL ICs.

Apparatus Required:-

Digital lab kit, single strand wires, breadboard, TTL IC's

Gates	IC NO.
AND	7408
OR	7432
NAND	7400
NOR	7402
NOT	7404
XOR	7486
XNOR	74266

Theory:-

Logic gates are idealized or physical devices implementing a Boolean function, which it performs a logical operation on one or more logical inputs and produce a single output. Depending on the context, the term may refer to an ideal logic gate, one that has for instance zero rise time and unlimited fan out or it may refer to a non-ideal physical device.

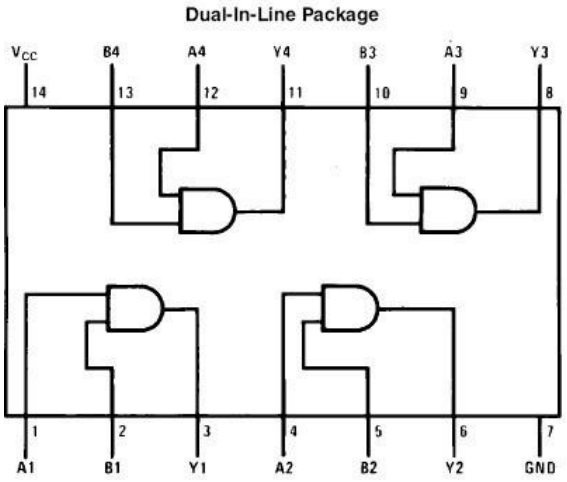
The main hierarchy is as follows:-

1. Basic Gates
2. Universal Gates
3. Advanced Gates

Basic Gates

1. **AND gate: -** Function of AND gate is to give the output true when both the inputs are true. In all the other remaining cases output becomes false. Following table justifies the statement:-

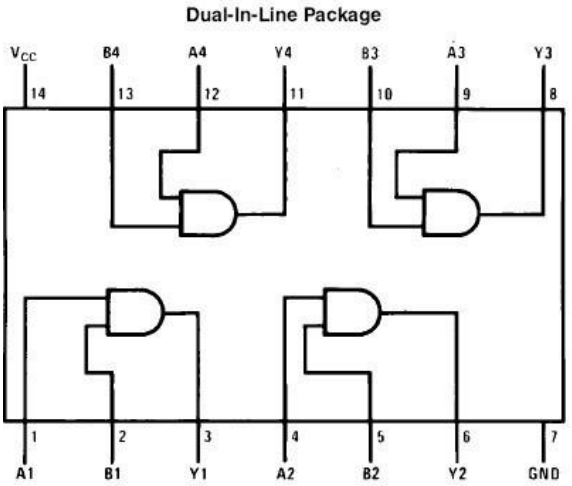
Input A	Input B	Output
0	0	0
0	1	0
1	0	0
1	1	1



IC 7408

2. **OR gate: -** Function of OR gate is to give output true when one of the either inputs are true .In the remaining case output becomes false. Following table justify the statement:-

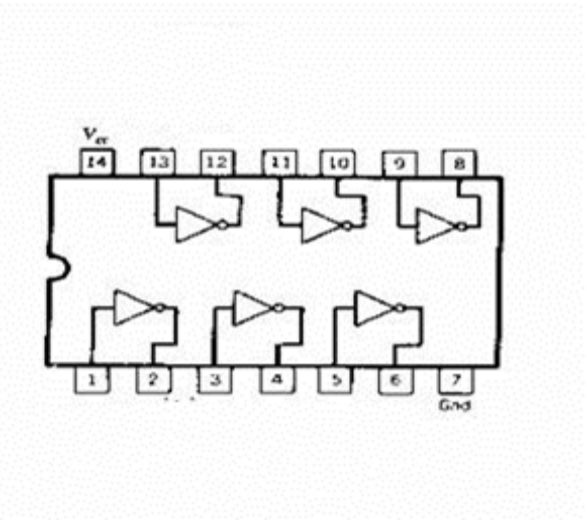
Input A	Input B	Output
0	0	0
0	1	1
1	0	1
1	1	1



IC 7432

3. **NOT gate: -** Function of NOR gate is to reverse the nature of the input .It converts true input to false and vice versa. Following table justifies the statement :-

Input	Output
1	0
0	1

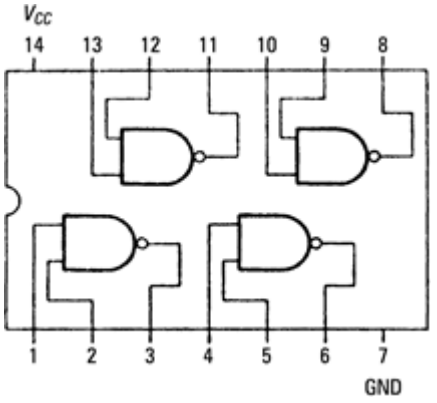


IC 7404

Universal Gates

1. **NAND gate: -** Function of NAND gate is to give true output when one of the two provided input are false. In the remaining output is true case. Following table justifies the statement :-

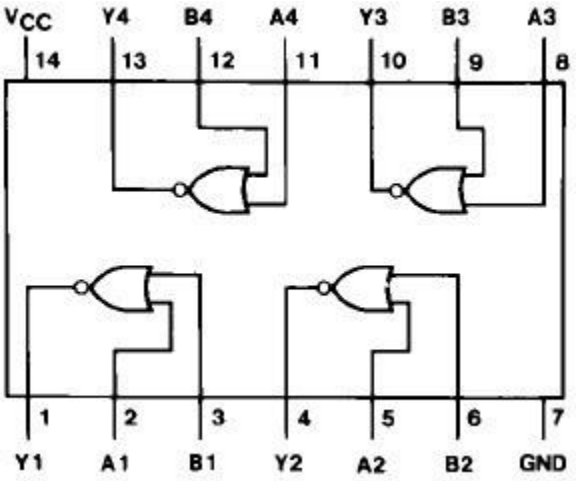
Input A	Input B	Output
0	0	1
0	1	1
1	0	1
1	1	0



IC 7400

2. **NOR gate: -** NOR gate gives the output true when both the two provided input are false. In all the other cases output remains false. Following table justifies the statement:-

Input A	Input B	Output
0	0	1
0	1	0
1	0	0
1	1	0

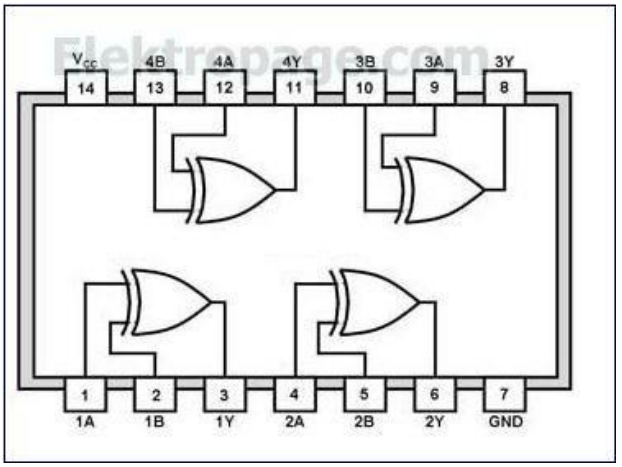


IC 7402

Advanced Gates

1. **XOR gate: -** The function of XOR gate is to give output true only when both the inputs are true. Following table explains this:-

Input A	Input B	Output
0	0	0
0	1	1
1	0	1
1	1	0



IC 74136

Procedure:-

- Place the breadboard gently on the observation table.
- Fix the IC which is under observation between the half shadow line of breadboard, so there is no shortage of voltage.
- Connect the wire to the main voltage source (V_{cc}) whose other end is connected to last pin of the IC (14 place from the notch).
- Connect the ground of IC (7th place from the notch) to the ground terminal provided on the digital lab kit.
- Give the input at any one of the gate of the ICs i.e. 1st, 2nd, 3rd, 4th gate by using connecting wires.(In accordance to IC provided).
- Connect output pins to the led on digital lab kit.
- Switch on the power supply.
- If led glows red then output is true, if it glows green output is false, which is numerically denoted as 1 and 0 respectively. The Color can change based on the IC manufacturer it's just verification of the Truth Table not the color change.

Result:-

All gates are verified. Observed output matches theoretical concepts.

Precautions:-

- All connections should be made neat and tight.
- Digital lab kits and ICs should be handled with utmost care.
- While making connections main voltage should be kept switched off.
- Never touch live and naked wires.

Pre Experiment Questions:-

1. What is a logic gate?

Ans: Logic gate is a physical device implementing a Boolean function and performs Logical operation on one or more logic inputs and produces a single logic output.

2. What are universal gates?

Ans: NAND and NOR gates are called universal gates as any type of logic gates or logic Functions can be implemented by these gates.

3. What are basic gates?

Ans: AND, OR, Not are called basic gates.

4. State De-Morgan's theorem.

Ans: $(x+y)^{l} = x^{l}y^{l}$

$(xy)^{l} = x^{l} + y^{l}$

5. What is the primary motivation for using Boolean algebra to simplify logic expressions?

Ans: (1) Boolean algebra reduces the number of inputs required.

(2) It will reduce number of gates

(3) It makes easier to understand the overall function of the circuit.

Post Experiment Questions:-

1. Which of the logical operations is represented by the + sign in Boolean algebra?

Ans: OR gate

2. Which of the two input logic gate can be used to implement an inverter circuit?

Ans: Ex-NOR gate

3. Which are the logic gates whose all output entries are logic 1 except for one entry there is logic 0?

Ans: NAND and NOR gate

4. TTL operates from a __5_____volt supply.

5. When the output of a NOR gate is high?

Ans : If all the inputs are low

EXPERIMENT - 2

FLIP FLOPS

Aim: - Verification of state tables of

1. R-S flip-flop
2. J - K flip-flop
3. T Flip-Flop
4. D Flip-Flop

Using NAND and NOR gates.

Apparatus: - IC 7400 (NAND Gate), IC 7402 (NOR Gate), IC 7408 (AND Gate).

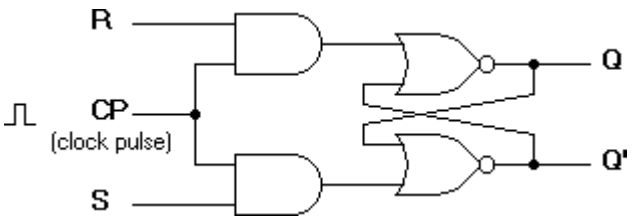
Theory: - In case of sequential circuits the effect of all previous inputs on the outputs is represented by a state of the circuit. Thus, the output of the circuit at any time depends upon its current state and the input. These also determine the next state of the circuit. The relationship that exists among the inputs, outputs, present states and next states can be specified by either the state table or the state diagram.

State Table:- The state table representation of a sequential circuit consists of three sections labelled present state next state and output. The present state designates the state of flip-flops before the occurrence of a clock pulse. The next state shows the states of flip-flops after the clock pulse, and the output section lists the value of the output variables during the present state.

Flip-Flop:-The basic one bit digital memory circuit is known as flip-flop. It can store either 0or 1. Flip-flops are classifieds according to the number of inputs.

R-S Flip-Flop:- The circuit is similar to SR latch except enable signal is replaced by clock pulse.

Logic Diagram

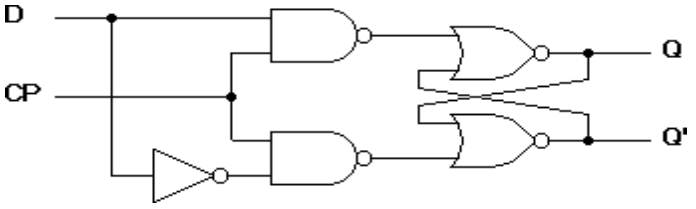


Characteristic table for S-R flip flop

Q S R	Q(t+1)
0 0 0	0
0 0 1	0
0 1 0	1
0 1 1	indeterminate
1 0 0	1
1 0 1	0
1 1 0	1
1 1 1	indeterminate

D Flip-Flop: -The modified clocked SR flip-flop is known as D-flip-flop. From the truth table of SR flip-flop we see that the output of the SR flip-flop is in unpredictable state when the inputs are same and high. In many practical applications, these input conditions are not required. These input conditions can be avoided by making then complement of each other.

Logic Diagram

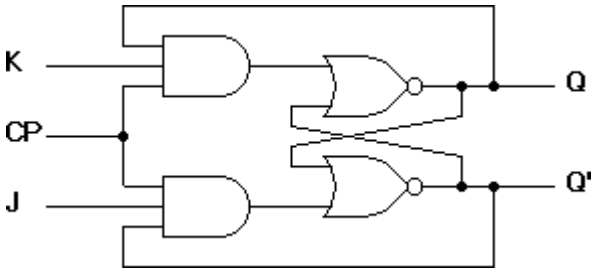


Characteristic table for D flip flop

Q D	Q(t+1)
0 0	0
0 1	1
1 0	0
1 1	1

J-K Flip-Flop:- In a RS flip-flop the input R=S=1 leads to an indeterminate output. The RS flip-flop circuit may be re-joined if both inputs are 1 than also the outputs are complement of each other.

Logic Diagram

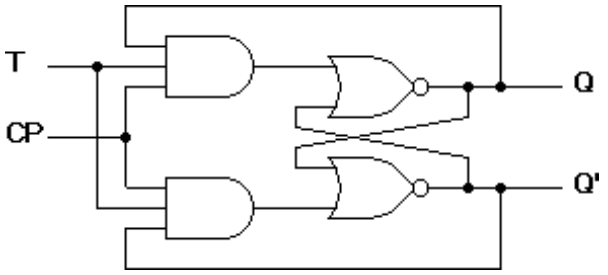


Characteristic table for J-K flip flop

Q J K	Q(t+1)
0 0 0	0
0 0 1	0
0 1 0	1
0 1 1	1
1 0 0	1
1 0 1	0
1 1 0	1
1 1 1	0

T Flip-Flop:- T flip-flop is known as toggle flip-flop. The T flip-flop is modification of the J-K flip-flop. Both the JK inputs of the JK flip-flop are held at logic 1 and the clock signal continuous to change.

Logic Diagram



Characteristic table for T flip flop

Q	T	Q(t+1)
0	0	0
0	1	1
1	0	1
1	1	0

Procedure:-

1. Connections are made as per circuit diagram.

2. Verify truth-tables for various combinations of input.

RESULT: - Study and verified truth-tables of various flip-flops.

Precaution:-

1. All the IC's should be checked before use the apparatus.
2. All LED's should be checked.
3. All connections should be tight.
4. Always connect GROUND first and then Vcc
5. The circuit should be off before change the connections .6.After completing the experiment switch off the supply to apparatus .

Pre Experiment Questions:-

1) What is a latch?

Ans: Storage elements that operate with signal levels are referred to as latches.

2) What is a flip flop?

Ans: Storage elements controlled by a clock transitions are called flip flop.

3) Differentiate between a latch and a flip flop.

Latch	Flip flop
1) A latch checks all its inputs continuously and changes its outputs accordingly at any time.	1) Flip flop samples its inputs and changes its outputs only at a time as determined by a clocking signal.
2) No clock is used	2) A clock is used

4) Differentiate between combinational and sequential circuits.

Combinational circuits.	Sequential circuits.
1) A circuit whose output is dependent only on the inputs at that instant	1) It depends on present and past history of the inputs
2) Easy to design	2) Harder to design
3) More expensive circuit	3) Cheaper circuit

5) What happens to the JK flip flop if the J input is treated as an inverter is wired between J and K inputs?

Ans: JK flip flop becomes D flip flop

Post Experiment Questions:-

1) How is a JK flip flop made to toggle?

Ans : If J=1 and K=1

2) When a JK flip flop is is in a no change condition?
   Ans: If J=0 and K=0

3) Which is the basic sequential building block in which the output follows the data input as long as the enable input is active?

   Ans: D latch

4) Define the characteristic equation of a flip flop.

Ans: The algebraic description of the next state of the flip flop is called characteristic equation.

5) How many stable states a flip flop has?

Ans: 2 stable states

EXPERIMENT -3

HALF ADDER AND HALF SUBSTRACTOR-BASIC GATES

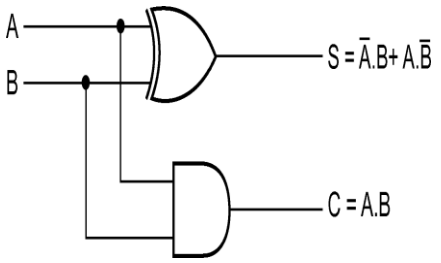
Aim 1: - To construct a Half adder circuit using basic gates

EQUIPMENTS / COMPONENTS REQUIRED :- Digital IC Trainer Kit, Ics – 7486, 7408

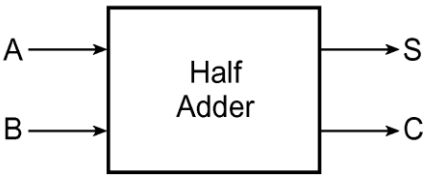
Theory:-
A half-adder is an arithmetic circuit block that can be used to add two bits. Such a circuit thus has two inputs that represent the two bits to be added and two outputs, with one producing the SUM output and the other producing the CARRY.
The Boolean expressions for the SUM and CARRY outputs are given by the equations

SUM S = A B + A B
CARRY C = A B

Procedure



A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1



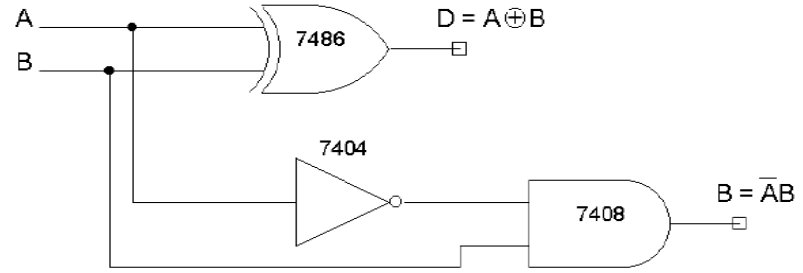
Aim 2: - To construct a Half subtracor circuit using basic gates

EQUIPMENTS / COMPONENTS REQUIRED :- Digital IC Trainer Kit, Ics – 7486, 7408,7404

Theory:-

The half subtractor is constructed using X-OR and AND Gate. The half subtractor has two input and two outputs. The outputs are difference and borrow. The difference can be applied using X-OR Gate, borrow output can be implemented using an AND Gate and an inverter.

Using X – OR and Basic Gates (a)Half Subtractor



INPUT		OUTPUT	
A	**B**	**BO**	**DIFF**
0	0	0	0
0	1	1	1
1	0	0	1
1	1	0	0

RESULT: - Study and verified truth-tables of Half adder and half substractor.

EXPERIMENT-4

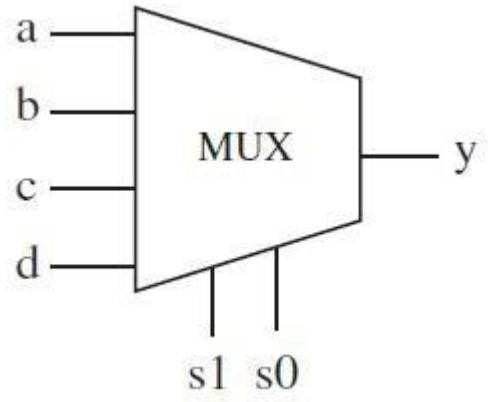
MULTIPLEXER

Aim:-Implementation of 4x1 multiplexer, using logic gates.

Apparatus: - Digital trainer kit, AND-7411, OR-7432, NOT-7404 Gate IC, Connecting wires.

Theory:-

MULTIPLEXER: - A multiplexer (MUX) is a device that accepts data from one of many input sources for transmission over a common shared line. To achieve this MUX has several data lines and a single output along with data-select inputs, which permit digital data on any of the inputs to be switched to the output line. The logic symbol for a 1 to 4 data selector/multiplexer is shown in Figure



Block diagram of multiplexer

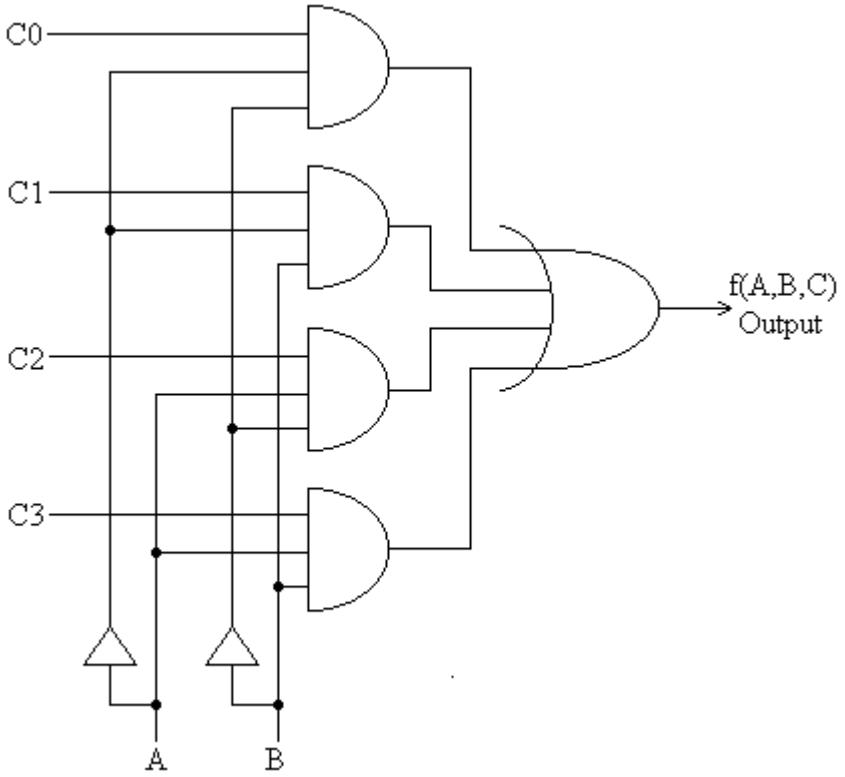
The selection lines decide the number of inputs lines of particular multiplexer. If the number of n inputs lines is equal to 2m, then m select lines are required to select one of the n input line.

Note that if a binary zero appears on the data-select lines then data on input line D0 will appear on the output. Thus, data output Y is equal to D0 if and only if S1=0 and S0=0.

Y=D0.S1’.S0’ Similarly, the data output is equal to D1, D2 and D3 for y=C1.S1’.S0’ ,Y=C2.S0’. S1 and Y=C3.S0.S1 respectively. Thus the total multiplexer logic expression, formed from ORing terms i.

The implementation of this equation is as shown in figure:

Logic diagram of multiplexer



DATA SELECT INPUT		INPUT SELECTED
S1	S0	
0	0	D0
0	1	D1
1	0	D2
1	1	D3

Truth table of mux

PROCEDURE:-

1 . Connections are made as per circuit diagram.
2. Verify the truth table.
3. Also connect Vcc and Ground then performed experiment.

Result: -Study of 4×1 multiplexer and verified its truth table.

Precautions:-

1. All ICs should be checked before starting the experiment.
2. All the connection should be tight.
3. Always connect ground first and then connect Vcc.
4. Suitable type wire should be used for different types of circuit.
5. The kit should be off before change the connections.
6. After completed the experiments switch off the supply of the apparatus

Pre Experiment Questions:-

1) What is a multiplexer?

Ans: A multiplexer accepts two or more streams of data and combines them into one stream.

2) How many select lines will a 16 to 1 multiplexer will have?

Ans: 4

3) What is the function of enable input on a multiplexer chip?

Ans: To activate the entire chip.

Post Experiment Questions:-

1) Why multiplexer is called as data selector?

Ans: Because multiplexer selects one of several input signals and directs to the output.

2) Give the applications of multiplexer.

Ans: Multiplexers are used in data routing, data selection, parallel to serial conversion and waveform generation.

3) How many control inputs are there in 16 to 1 demultiplexer?

Ans: 4

4) How many select lines will a 32:1 multiplexer will have?

Ans: For 32 inputs, 5 select lines will be required, as $2^5 = 32$.

EXPERIMENT-5

DE-MULTIPLEXER

Aim:-Implementation of 1x4 De-multiplexer, using logic gates.

Apparatus: - Digital trainer kit, AND-7411, , NOT-7404 Gate IC, Connecting wires.

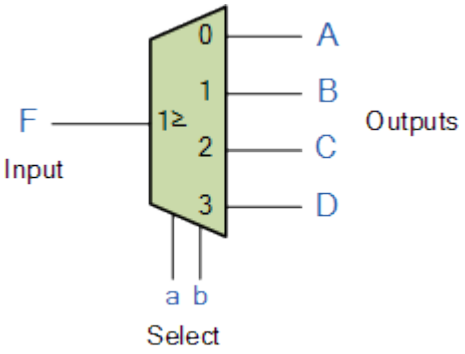
Theory:-

DE-MULTIPLEXER: -

The function of Demultiplexer is in contrast to multiplexer function. It takes information from one line and distributes it to a given number of output lines. For this reason, the demultiplexer is also known as a data distributor. Decoder can also be used as demultiplexer. In the 1: 4 demultiplexer circuit, the data input line goes to all of the AND gates. The data select lines enable only one gate at a time and the data on the data input line will pass through the selected gate to the associated data output line.

The logic symbol for a 1 to 4 data selector/multiplexer is shown in Figure

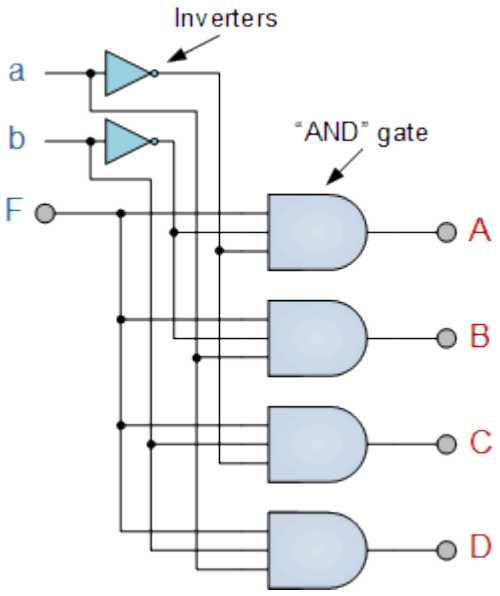
Block diagram of multiplexer



The implementation of this equation is as shown in figure:

s

Logic diagram of multiplexer



TRUTH TABLE

INPUT	SELECT LINE		OUTPUT			
F	**a**	**b**	**A**	**B**	**C**	**D**
1	0	0	1	0	0	0
1	0	1	0	1	0	0
1	1	0	0	0	1	0
1	1	1	0	0	0	1

PROCEDURE:-

1 . Connections are made as per circuit diagram.
4. Verify the truth table.
5. Also connect Vcc and Ground then performed experiment.

Result: -Study of 1×4 De multiplexer and verified its truth table.

EXPERIMENT-6

BINARY TO GRAY CONVERSION

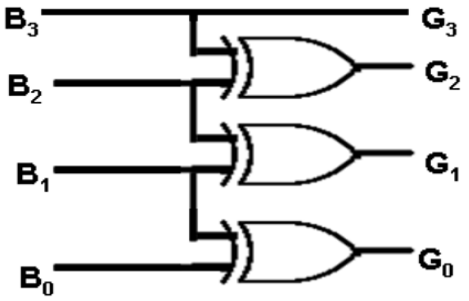
Aim: - To convert binary numbers to gray codes.

EQUIPMENTS / COMPONENTS REQUIRED :- Digital IC Trainer Kit, and IC 7486

Theory

A given binary number can be converted into its Gray code equivalent by going through the following steps:

1. Begin with the most significant bit (MSB) of the binary number. The MSB of the Gray code equivalent is the same as the MSB of the given binary number.
2. The second most significant bit, adjacent to the MSB, in the Gray code number is obtained by adding the MSB and the second MSB of the binary number and ignoring the carry, if any. That is, if the MSB and the bit adjacent to it are both '1', then the corresponding Gray code bit would be a '0'.
3. The third most significant bit, adjacent to the second MSB, in the Gray code number is obtained by adding the second MSB and the third MSB in the binary number and ignoring the carry, if any.
3. The process continues until we obtain the LSB of the Gray code number by the addition of the LSB and the next higher adjacent bit of the binary number.
4.
5. $B_3 = G_3$
6. $B_2 \oplus B_3 = G_2$
7. $B_1 \oplus B_2 = G_1$
8. $B_0 \oplus B_1 = G_0$



Inputs				Outputs			
B3	B2	B1	B0	G3 (V)	G2 (V)	G1 (V)	G0 (V)
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	1
0	0	1	1	0	0	1	0
0	1	0	0	0	1	1	0
0	1	0	1	0	1	1	1
0	1	1	0	0	1	0	1
0	1	1	1	0	1	0	0
1	0	0	0	1	1	0	0
1	0	0	1	1	1	0	1
1	0	1	0	1	1	1	1
1	0	1	1	1	1	1	0
1	1	0	0	1	0	1	0
1	1	0	1	1	0	1	1
1	1	1	0	1	0	0	1
1	1	1	1	1	0	0	0

GRAY TO BINARY CONVERSION

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The Most Significant Bit (MSB) of the binary code is always equal to the MSB of the given binary number.

2.Other bits of the output binary code can be obtained by checking gray code bit at that index. If current gray code bit is 0, then copy previous binary code bit, else copy invert of previous binary code bit.

There are four inputs and four outputs. The input variable are defined as G_3, G_2, G_1, G_0 and the output variables are defined as B_3, B_2, B_1, B_0. From the truth table, combinational circuit is designed.The logical expressions are defined as :

$$G_0 \oplus G_1 \oplus G_2 \oplus G_3 = B_0$$

$$G_1 \oplus G_2 \oplus G_3 = B_1$$

$$G_2 \oplus G_3 = B_2$$

$$G_3 = B3$$

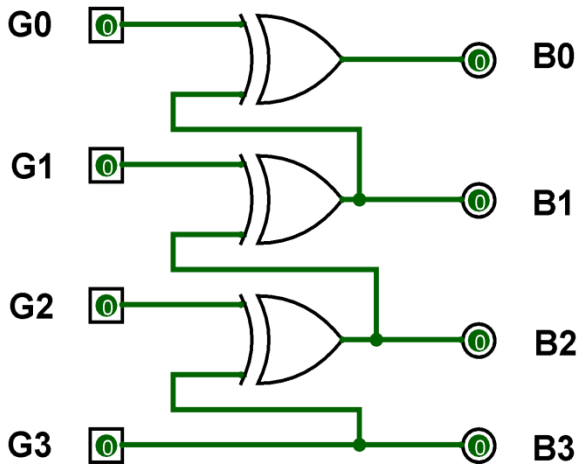


Figure-3: Gray to Binary Code Converter Circuit

Gray code				Natural-binary code			
G3	G2	G1	G0	B3	B2	B1	B0
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	1
0	0	1	1	0	0	1	0
0	1	0	0	0	1	1	1
0	1	0	1	0	1	1	0
0	1	1	0	0	1	0	0
0	1	1	1	0	1	0	1
1	0	0	0	1	1	1	1
1	0	0	1	1	1	1	0
1	0	1	0	1	1	0	0
1	0	1	1	1	1	0	1
1	1	0	0	1	0	0	0
1	1	0	1	1	0	0	1
1	1	1	0	1	0	1	1
1	1	1	1	1	0	1	0

Figure-4: Gray to Binary Code Converter Truth Table

EXPERIMENT-7

BASIC GATES USING UNIVERSAL GATES

AIM: Implement basic gates using universal gates

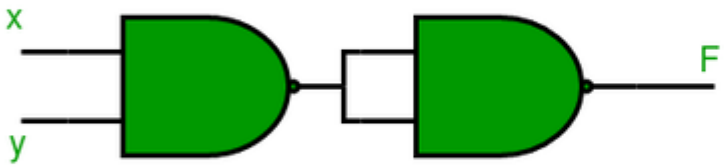
Components required : IC 7400,7402,Trainer kit

Theory: In Boolean Algebra, the **NAND** and **NOR** gates are called **universal gates** because any digital circuit can be implemented by using any one of these two i.e. any logic gate can be created using NAND or NOR gates only.

1. Implementation of AND Gate using Universal gates.

a) Using NAND Gates

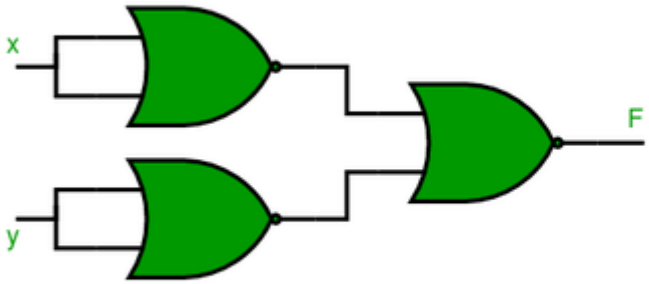
The AND gate can be implemented by using two NAND gates in the below fashion:



x	y	F
0	0	0
0	1	0
1	0	0
1	1	1

b) Using NOR Gates

Implementation of AND gate using only NOR gates as shown below:

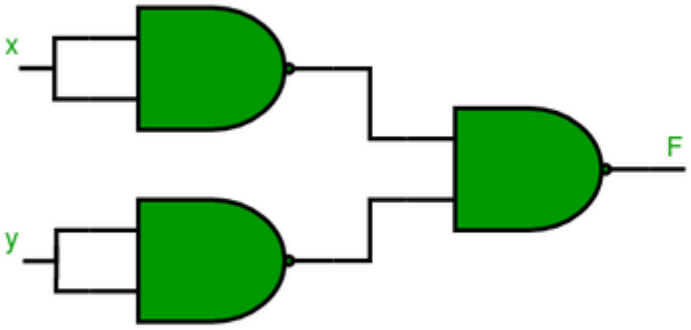


x	y	F
0	0	0
1	0	0
1	1	1
0	1	0

Implementation of OR Gate using Universal gates.

a) Using NAND Gates

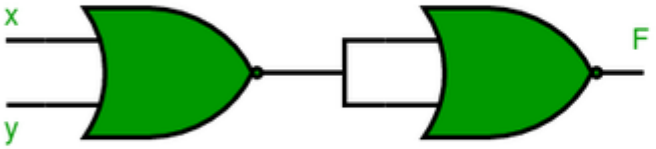
The OR gate can be implemented using the NAND gate as below:



x	y	F
0	0	0
0	1	1
1	0	1
1	1	1

b) Using NOR Gates

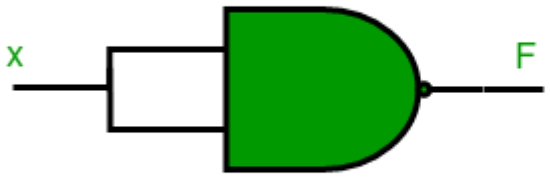
Implementation of OR gate using two NOR gates as shown in the picture below:



x	y	F
0	0	0
0	1	1
1	0	1
1	1	1

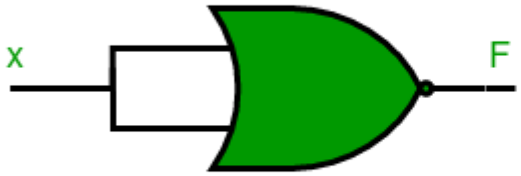
3. Implementation of NOT Gate using Universal gates.

a) Using NAND Gates



x	F
0	1
1	0

b) Using NOR Gates



x	F
0	1
1	0

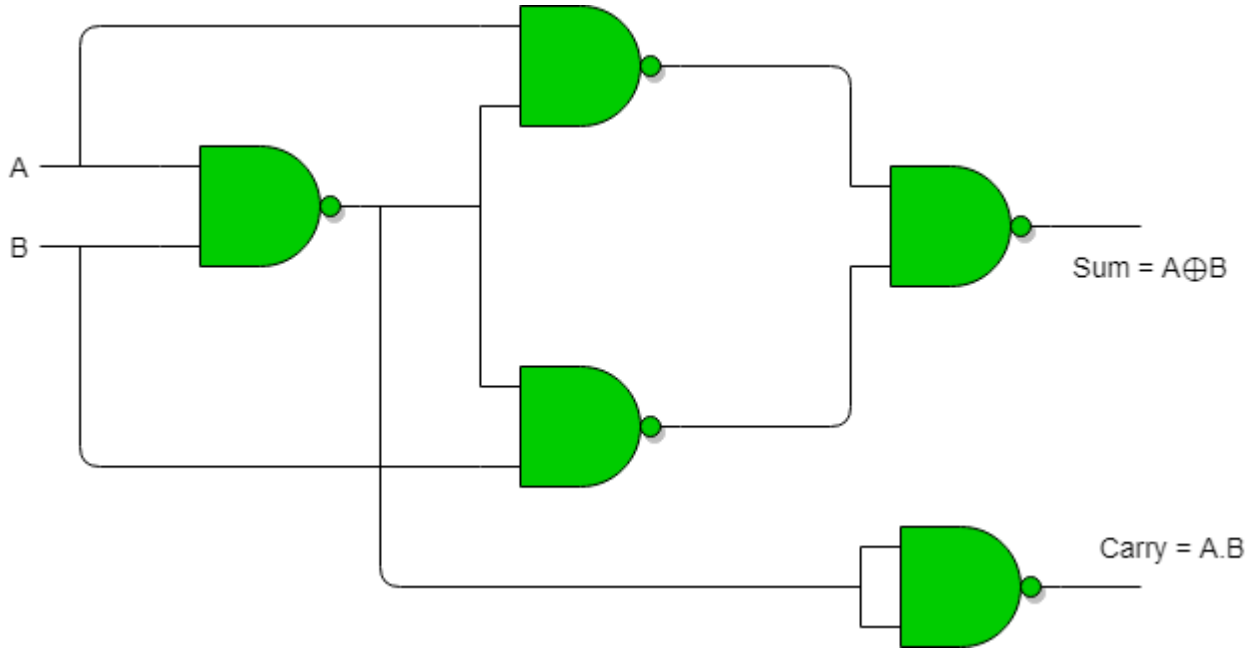
EXPERIMENT-8

HALF ADDER AND HALF SUBSTRACTOR-UNIVERSAL GATES

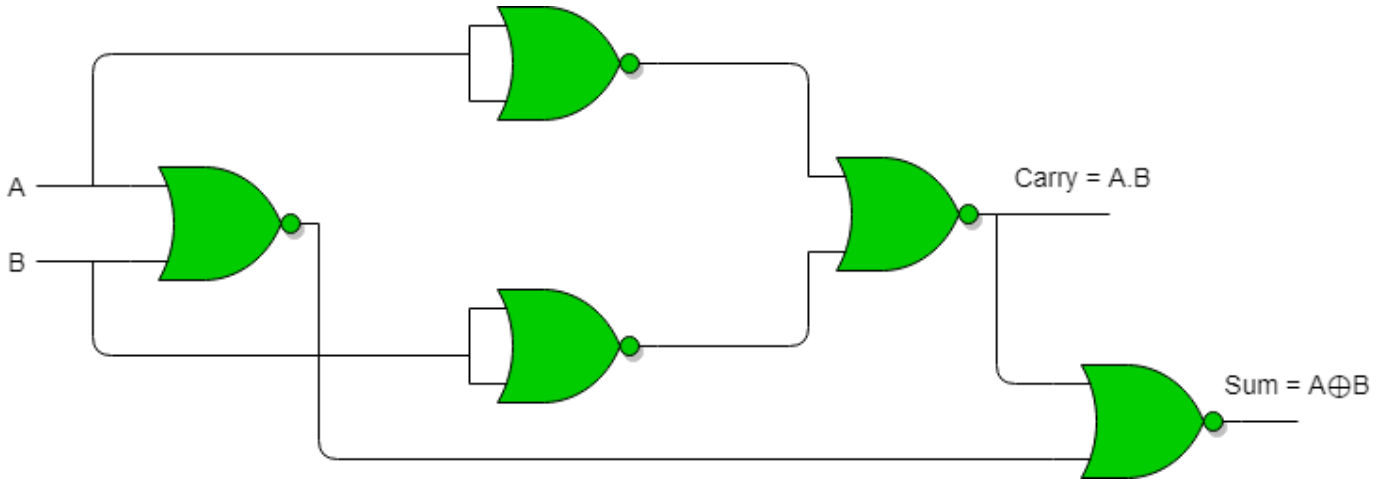
AIM: Implement half adder and half substractor using NAND and NOR gates

COMPONENTS REQUIRED: IC 7400,7402

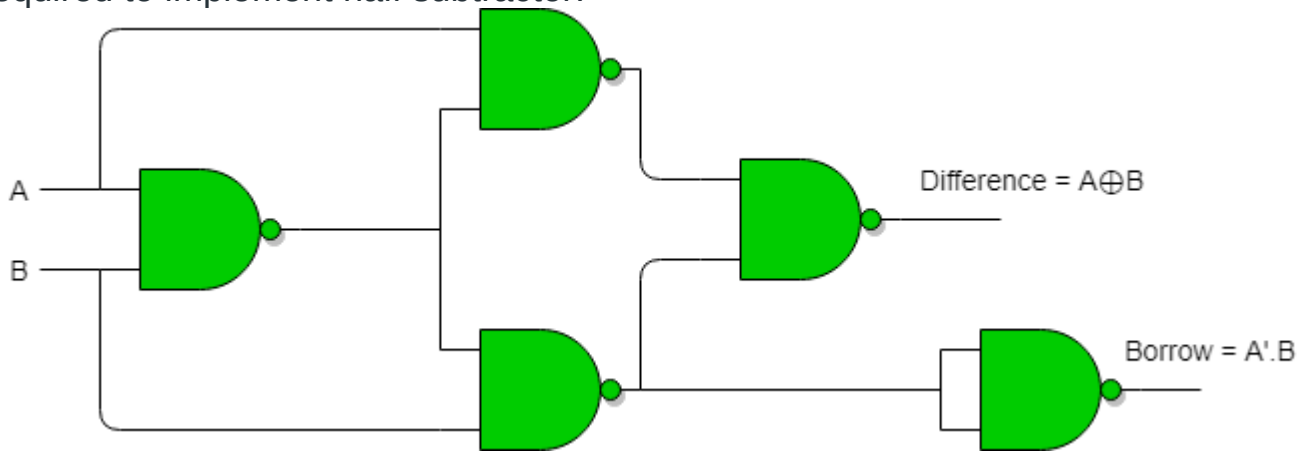
THEORY: Universality: NAND and NOR gates are considered universal gates because they can be used to implement any logical function, including binary arithmetic functions such as addition and subtraction.
Implementation of Half Adder using NAND gates : Total 5 NAND gates are required to implement half adder.



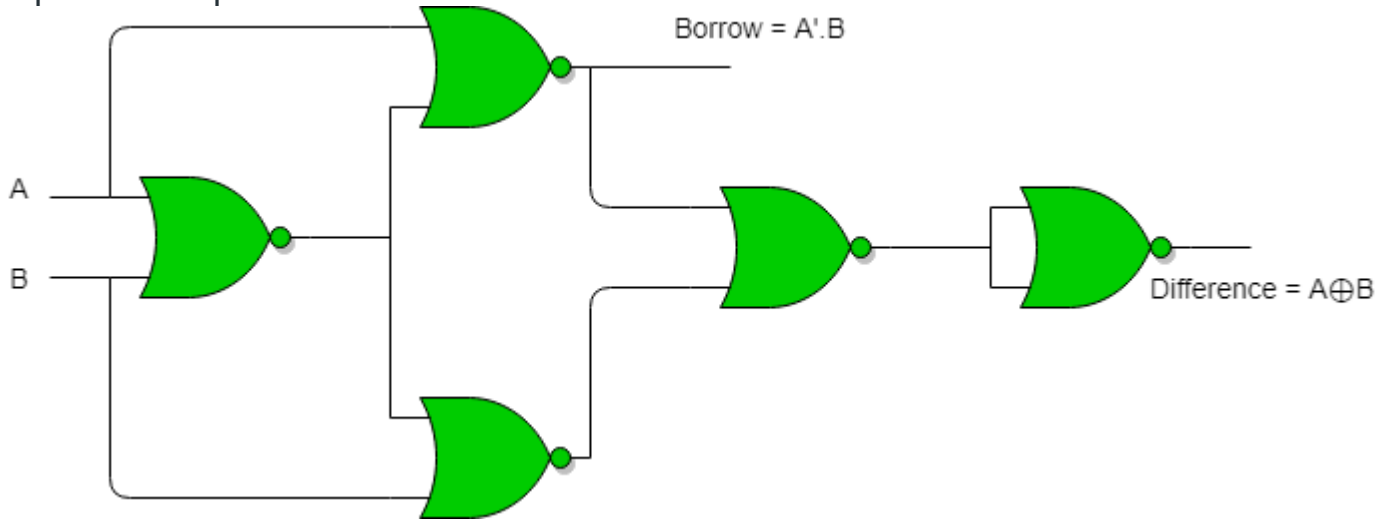
Implementation of Half Adder using NOR gates : Total 5 NOR gates are required to implement half adder.



Implementation of Half Subtractor using NAND gates : Total 5 NAND gates are required to implement half subtractor.



Implementation of Half Subtractor using NOR gates : Total 5 NOR gates are required to implement half subtractor.



EXPERIMENT-9

MULTIPLEXER USING IC 74153

AIM: To implement multiplexer using IC74153

COMPONENTS REQUIRED: Digital trainer kit ,IC 74153

THEORY:

The integrated circuit **74153** contains **two 4-way multiplexers** with common **A** and **B** selection inputs. Each multiplexer has a validation input **G** (**STROBE**). This, brought to **state 1**, forces the output of the multiplexer corresponding to **state 0** independently of the state of the other inputs.The pinout and the logic diagram of this integrated circuit are given in Figure 32, while Figure 33 gives its truth table.

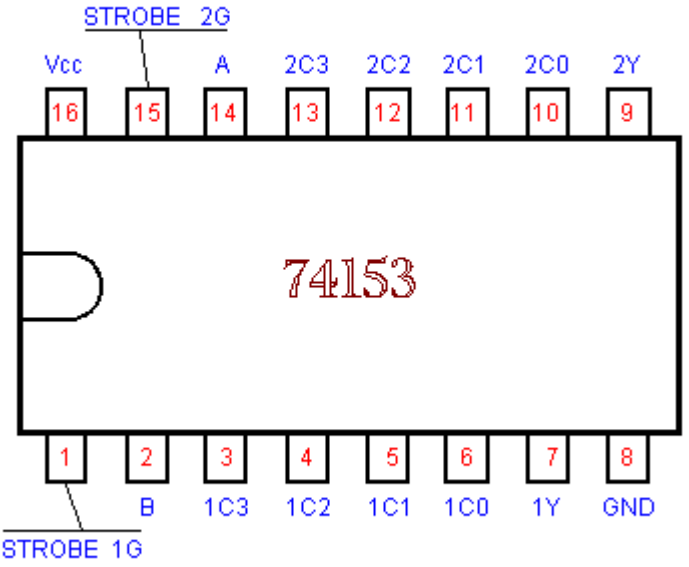


Fig. 32. - Brochage du circuit intégré 74153.

EXPERIMENT-10

DECADE COUNTER

Aim:to construct decade counter using IC 7490

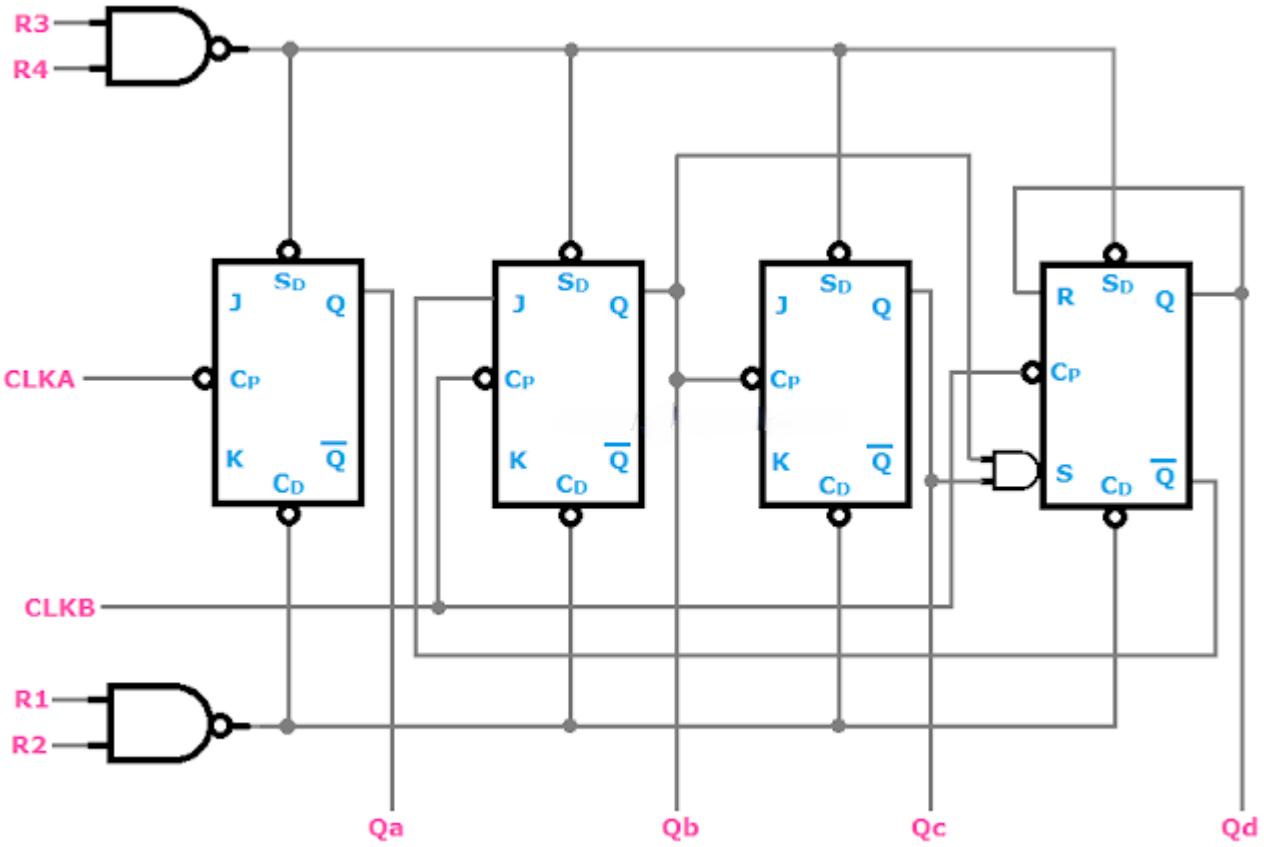
COMPONENTS REQUIRED:IC 7490,trainer kit

Theory:

Decade counter using IC 7490

The IC 7490, often referred to as a "Decade Counter," is an integrated circuit used in digital electronics for counting and sequencing purposes. Developed as a monolithic device, the IC 7490 contains a combination of flip-flops and digital logic gates that allow it to count in a variety of ways. This versatile IC has found applications in numerous electronic systems and is especially useful in digital counting, frequency division, and timing applications.

CIRCUIT DAIGROM



TRUTH TABLE

Count	Qd	Qc	Qb	Qa
(Start) 0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
(New Cycle) 10	0	0	0	0

.

EXPERIMENT-11

De-Morgan's Theorem

AIM: Implement a logic circuit to verify demorgans theorem.

COMPONENTS REQUIRED:Trainer kit

THEORY:

De Morgan's Law Boolean Algebra defines the relation between the OR, AND, and the complements of variables, and is given for both the complement of AND and OR of two values. In Boolean Algebra there are two De Morgan's Laws that are:

- First De Morgan's Law
- Second De Morgan's Law

Let's understand these laws in detail as below:

First De Morgan's Law in Boolean Algebra

First De Morgan's law states that **"The complement of OR of two or more variables is equal to the AND of the complement of each variable."**
Let A and B be two variables, then mathematically First De Morgan's Law is given as:

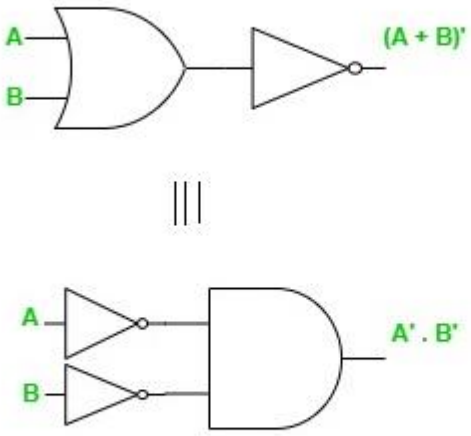
$$(A + B)' = A' . B'$$

Where

- *+ represents the OR operator between variables,*
- ***.** represents AND operator between variables, and*
- ***'** represents complement operation on variable.*

First De Morgan's Law Logic Gates

In context to logic gates and Boolean Algebra, De Morgan's Law states that "Both the logic gate circuits i.e., NOT gate is added to the output of OR gate, and NOT gate is added to the input of AND gate, are equivalent. These two logic gate circuits are given as follows:



First De Morgan's Law Truth Table

The truth table for first De Morgan's Law is given as follows:

A	B	A + B	(A + B)'	A'	B'	A'. B'
0	0	0	1	1	1	1
0	1	1	0	1	0	0
1	0	1	0	0	1	0
1	1	1	0	0	0	0

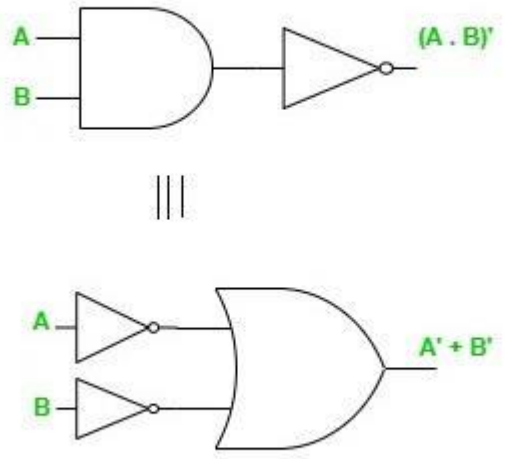
2 .Second De Morgan's law states that **"The complement of AND of two or more variables is equal to the OR of the complement of each variable."**
Let A and B be two variables, then mathematically Second De Morgan's Law is given as:

$$(A \cdot B)' = A' + B'$$

Second De Morgan's Law Truth Table

The truth table for the second De Morgan's Law is given as follows:

A	B	A . B	(A. B)'	A'	B'	A' + B'
0	0	0	1	1	1	1
0	1	0	1	1	0	1
1	0	0	1	0	1	1
1	1	1	0	0	0	0



RESULT: Verified demorgans theorem successfully.